

Please attach this cover sheet to your completed homework

Homework 2 (Ch. 8)

ECE 311 – Hang Gao

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Problem	Course outcome	Grade
8.1	1.c.	/36
8.2	1.c.	/20
8.5	1.c.	/15
8.6	1.c.	/15
8.11	1.c.	/14
TOTAL:		/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #2

Due date: Sunday, February 8th by 11:59 PM (Electronically – Canvas)

8.1 A Y-connected balanced three-phase source is feeding a balanced three-phase load. The voltage and current of the source coil are

$$v(t) = 340 \sin(377t + 0.5236)V$$

$$i(t) = 100 \sin(377t + 0.87266)A$$

The phase shift angles in the two equations are in radians. Calculate the following:

- The rms phase voltage
- The rms line-to-line voltage
- The rms current in the source
- The rms current in the transmission line
- The frequency of the supply
- The power factor at the source side, state leading or lagging
- The three-phase real power delivered to the load
- The three-phase reactive power delivered to the load
- If the load is connected in delta configuration, calculate the load impedance

8.2 The current and voltage of a wye-connected load are:

$$\bar{V}_{ca} = 480 \angle -60^\circ V$$

$$\bar{I}_b = 20 \angle 120^\circ A$$

- Compute $\bar{V}_{an}$
- Compute $\bar{I}_a$
- Compute the power factor angle
- Compute the real power of the load

8.5 A balanced wye-connected load of $(4+j3)\Omega$ is connected across a three-phase source of 173 V (line-to-line).

- Find the transmission line current
- Find the power factor, the complex power, the real power, and the reactive power of the load.
- With V_{ab} as the reference, sketch the phasor diagram that shows all voltages and currents.

- 8.6** Two three-phase wye-connected loads are in parallel across a three-phase supply. The first load draws a current of 20 A at 0.9 power factor leading, and the second load draws a current of 30 A at 0.8 power factor lagging. Calculate the following:
- The transmission line current
 - The power factor of the load
 - The real power supplied by the source if the line-to-line voltage of the transmission line is 400 V

8.11 A three-phase wye-connected source is energizing a delta-connected load. The phase voltage of the transmission line is $\bar{V}_{bn} = 120 \angle 0^\circ \text{V}$, and the total load impedance is $\bar{Z} = 9 \angle 30^\circ \Omega$. Compute the line current $\bar{I}_a$ and the real power consumed by the delta load.

8.1 **Note:** the following assumes that, unless otherwise notated, voltages and current magnitudes are given in rms values. Additionally, when calculating voltage, current, and impedance in *phasor* form, $\bar{V}_{an}$ is used as the reference voltage; that is, it is assumed to have a phase angle $\theta = 0^\circ$. Finally, I have assumed throughout this problem that the wye-connected source is powering a delta-connected load.

- a) For a sinusoidal voltage in a three-phase system given in the form $V_{\max} \sin(\omega t \pm \theta)$, the rms phase voltage V_{ph} is given by

$$V_{ph} = \frac{V_{\max}}{\sqrt{2}} = \frac{340}{\sqrt{2}} = 240.42 \text{ V.}$$

- b) The line-to-line rms voltage in a three-phase, balanced, wye-connected source may be found using the phase voltages, where

$$\begin{aligned}\bar{V}_{an} &= V_{ph} \angle 0^\circ \\ \bar{V}_{bn} &= V_{ph} \angle -120^\circ \\ \bar{V}_{cn} &= V_{ph} \angle 120^\circ,\end{aligned}$$

and

$$\begin{aligned}\bar{V}_{ab} &= \sqrt{3}V_{ph} \angle 30^\circ \\ \bar{V}_{bc} &= \sqrt{3}V_{ph} \angle -90^\circ \\ \bar{V}_{ca} &= \sqrt{3}V_{ph} \angle 150^\circ.\end{aligned}$$

Therefore, solving just for the rms value of the line-to-line voltages,

$$V_{ab} = V_{bc} = V_{ca} = \sqrt{3}V_{ph} = 416.413 \text{ V.}$$

- c) The rms source current through a three-phase, wye-connected, balanced source, I_a may be found as

$$\frac{I_{\max}}{\sqrt{2}} = \frac{100}{\sqrt{6}} = 70.711 \text{ A}$$

- d) The rms value of the line current for any phase connected to a three-phase balanced load are found using a similar formula to phase voltage,

$$I_a = I_b = I_c = \frac{I_{\max}}{\sqrt{2}} = \frac{100}{\sqrt{2}} = 70.711 \text{ A.}$$

- e) The source's angular frequency in a three-phase balanced system is the same in all three lines, given in the time-dependent formula for voltage and current as ω , in radians per second. Converting $\omega = 377 \frac{\text{rad}}{\text{s}}$ to units of frequency f in hertz,

$$f = \frac{\omega}{2\pi} = \frac{377}{2\pi} = 60 \text{ Hz.}$$

- f) Power factor may be found as the cosine of the phase angle difference between voltage and current, $\theta_{VI} = \theta_V - \theta_I$,

$$P.F. = \cos(\theta_{VI}) = \cos(-0.3491) = 0.9397 \text{ (leading),}$$

since the phase angle difference θ_{VI} is negative, implying a capacitive load and a leading current waveform.

- g) Three-phase real power $P = 3P_{ph} = 3V_{ph}I_{ph} \cos(\theta_{VI})$, so

$$P = (3 \times 240.416 \times 70.711) \cos(-0.3491) = 47.924 \text{ kW.}$$

- h) Three-phase reactive power $Q = 3P_{ph} \sin(\theta_{IV})$, so

$$Q = (3 \times 240.416 \times 70.711) \sin(-0.3491) = -17.445 \text{ kVar.}$$

- i) Given a delta-connected three-phase balanced load, the load impedance $\bar{Z}$ may be found using

$$\bar{Z} = \frac{\bar{V}_{ab}}{\bar{I}_{ab}} = \frac{\bar{V}_{bc}}{\bar{I}_{bc}} = \frac{\bar{V}_{ca}}{\bar{I}_{ca}}.$$

Finding $\bar{I}_{ab}$, we may use the relationship

$$\bar{I}_a = \sqrt{3} \bar{I}_{ab} \angle -30^\circ \Rightarrow \bar{I}_{ab} = \frac{70.11}{\sqrt{3}} \angle 50^\circ = 40.83 \angle 50^\circ \text{ A.}$$

Then,

$$\bar{Z} = \frac{416.41 \angle 30^\circ}{40.83 \angle 50^\circ} = 10.20 \angle -20^\circ \Omega.$$

- 8.2 a) $\bar{V}_{an}$ may be found as

$$\bar{V}_{an} = V_{ph} \angle 0^\circ.$$

Thus, finding $\bar{V}_{an}$,

$$\sqrt{3} V_{ph} \angle 150^\circ = \bar{V}_{ca} \Rightarrow \bar{V}_{an} = \frac{480 \angle -60^\circ}{\sqrt{3} \angle 150^\circ} = 277.13 \angle 150^\circ \text{ V.}$$

- b) $\bar{I}_a$ may be found as

$$\bar{I}_a = I_b \angle (\theta_{I_b} + 120^\circ) = 20 \angle -120^\circ \text{ A.}$$

- c) The phase angle difference $\theta_{VI} = -90^\circ$, so

$$P.F. = \cos(-1.5708) = 0 \text{ (leading),}$$

implying a purely capacitive load.

- d) Real power P may be found as

$$P = 3P_{ph} \cos(\theta_{VI}) = (3 \times 277.128 \times 20) \cos(-90^\circ) = 0 \text{ W.}$$

- 8.5 a) Using $\bar{V}_{ab}$ as the reference voltage, with a phase angle of $\theta = 0^\circ$, we may find the voltage across the load as

$$\bar{V}_{an} = \frac{V_{ab}}{\sqrt{3}} \angle -30^\circ = 99.88 \angle -30^\circ \text{ V,}$$

and the transmission line current as

$$\bar{I}_a = \frac{\bar{V}_{an}}{Z} = \frac{99.88 \angle -30^\circ}{5 \angle 36.87^\circ} = 19.98 \angle -66.87^\circ \Omega$$

- b) Given a phase angle difference $\theta_{VI} = 66.87^\circ$,

$$P.F. = \cos(1.1671) = 0.3928 \text{ (lagging),}$$

$$S = P + jQ = 3V_{an}I_a = 3 \times 99.88 \times 19.98 = 5.987 \text{ kVA,}$$

$$P = S \cos(\theta_{VI}) = [5.987 \times 10^3] \cos(1.202) = 2.352 \text{ kW,}$$

$$Q = S \sin(\theta_{VI}) = [5.987 \times 10^3] \sin(1.202) = 5.506 \text{ kVar.}$$

- c) The following diagram was generated in MATLAB:

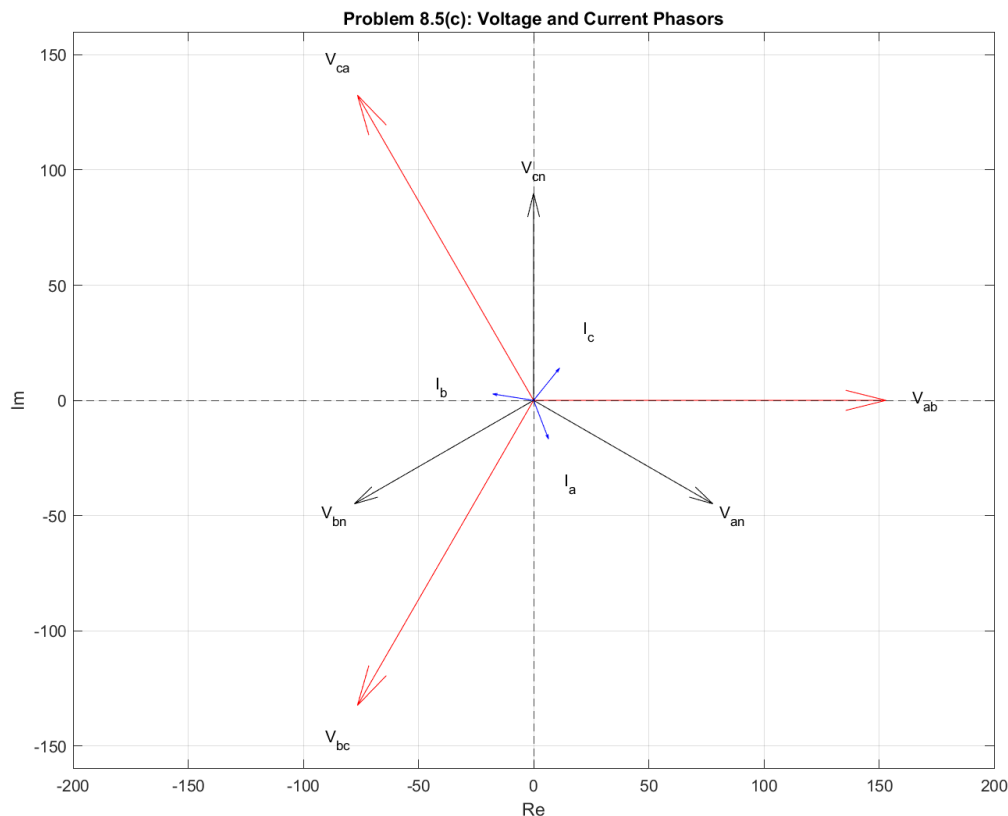


Figure 1: Voltage and Current Phasors

- 8.6 a) The total current through one phase A of the transmission line can be derived using KCL, and simply adding the current through the parallel loads together,

$$\bar{I}_a = 20\angle 25.84^\circ + 30\angle -36.87^\circ = 43.01\angle -12.46^\circ \text{ A.}$$

Then, the other two phases can be derived by simply shifting their phase angles 120° backwards,

$$\bar{I}_a = 43.01\angle -12.46^\circ \text{ A}$$

$$\bar{I}_b = 43.01\angle -132.46^\circ \text{ A}$$

$$\bar{I}_c = 43.01\angle 107.54^\circ \text{ A.}$$

- b) Assuming a reference voltage phase angle of $\theta = 0^\circ$ for V_{an} , we may derive the power factor as

$$P.F. = \cos(\theta_{VI}) = \cos(0.2175) = 0.9764 \text{ (lagging).}$$

- c) Using the phase voltage $V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$, we may find real power as

$$P = [3 \times 230.94 \times 43.01] \cos(0.2175) = 29.096 \text{ kW.}$$

- 8.11 Given $\bar{V}_{bn} = 120\angle 0^\circ \text{ V}$, $\bar{V}_{an} = 120\angle 120^\circ \text{ V}$, and $\bar{V}_{ab} = 207.85\angle 150^\circ \text{ V}$. Then,

$$\bar{I}_{ab} = \frac{\bar{V}_{ab}}{\bar{Z}} = \frac{207.85\angle 150^\circ}{9\angle 30^\circ} = 23.09\angle 120^\circ \text{ A.}$$

Converting load current to transmission line current,

$$\bar{I}_a = \sqrt{3} \bar{I}_{ab} \angle -30^\circ = \sqrt{3} \times 23.09 \angle 90^\circ = 39.99\angle 90^\circ \text{ A.}$$

Using the load current and voltage along with the phase angle difference to calculate real power consumed by the load,

$$3I_{ab}V_{ab} \cos(0.5236) = 12.469 \text{ kW.}$$